Tuorui Peng

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Summary

Tuorui Peng is a statistics Ph.D. candidate at Northwestern University (expected Jun. 2028) with experience in statistical machine learning, Python/R algorithm implementation, applied machine learning, and large-scale data pipelines. Seeking Summer 2027 internships across data science, applied ML/AI, and other quantitative technical roles.

Education

Doctor of Philosophy in Statistics — **Expected Jun. 2028**
Northwestern University · Advisor: Prof. Matey Neykov. — *Evanston, IL, USA*

M.S. in Statistics — **Jun. 2025**
Northwestern University · awarded en route — *Evanston, IL, USA*

B.S. in Mathematics and Physics, Minor in Statistics — **Jun. 2023**
Tsinghua University — *Beijing, China*

Experience

Efficient Monotone Single-Index Estimation Beyond Elliptical Designs
Northwestern University · Oct. 2025 – Present

- Designed a new algorithm that overcomes the distributional limitations of prior algorithms for monotone single-index estimation. Implemented data-driven adaptive parameter selection.
- Built cross-language packages (R version + Python version over one shared native core, BLAS-accelerated kernels, unit-tested).
- Experiments over various complicated designs show a wide margin in recovering the index direction over established rank-correlation and dimension-reduction baselines.

Robust Inference under Shape Constraints with Heavy-Tailed Noise
Northwestern University · Jan 2025 – April 2026

- Developed theory and minimax-optimal guarantees for robust mean estimation under heavy-tailed and adversarially contaminated noise with nonconvex (star-shaped) constraints; preprint under review at *Bernoulli* (arXiv:2604.05063).
- Follow-up work on the behavior near the breakdown point, exhibiting an interaction between the adversary and the geometry of the constraint set.

Distribution-Free Inference & NN Modeling for Gene Expression
Tsinghua University · Dec. 2022 – May 2023

- Built an end-to-end Python pipeline for genotype simulation, Mixture Density Network training, and KL-divergence-based evaluation against the known true conditional law.
- Extended conformal prediction to CDF-valued targets via a Wasserstein-1 score, yielding distribution-free bands with valid 90% / 95% / 99% coverage; validated over data-size, dimensionality, causal-rate and error-model settings.

PMC-Patients — Large-Scale Patient Summary Dataset & Retrieval Benchmark
Tsinghua University · Dec. 2021 – Jul. 2022

- Built components of the data-collection pipeline behind a peer-reviewed clinical dataset, converting PubMed Central case reports into structured patient records (section-level triggers, paragraph-level extractors, regex demographic parsing); a large scale dataset: 167k patient summaries, 3.1M patient–article and 293k patient–patient annotations.
- Fine-tuned BERT-based dense retrieval encoders and benchmarked them against baselines.
- Evaluated on a held-out 5k-query split, over a corpus of 11.7M articles and 155k candidate patients; published in *Scientific Data* (Nature Portfolio).

Skills, Teaching & Honors

Programming: Python, R, LaTeX, Mathematica.
ML & Data: PyTorch, scikit-learn, NumPy, pandas; web scraping, text mining, dataset construction, retrieval benchmarking.
Teaching: Teaching Assistant, Northwestern University — Linear Regression, Statistical Inference, Experimental Design, High-Dimensional Statistics.
Honors: Honorable Mention, MCM (2022); Scholarship for Academic Advance, Tsinghua Univ. (2020).
Languages: English (fluent), Chinese (native), Cantonese (fluent).

Publications

Peng, T., Prasadan, A., & Neykov, M. (2026). Robust mean estimation under star-shaped constraints with heavy-tailed noise. Under review at *Bernoulli*. Preprint: arXiv:2604.05063.
Zhao, Z., Jin, Q., Chen, F., Peng, T., & Yu, S. (2023). A large-scale dataset of patient summaries for retrieval-based clinical decision support systems. *Scientific Data*, 10(1), 909.